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Dual-axis attention network for granular load forecasting in industrial and residential demand response

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ABSTRACT

Smart grids require precise demand response management through accurate granular-level load forecasting capabilities. This paper introduces a novel Dual-Axis Attention Network (DAAN) that addresses the critical challenge of short-term load forecasting at both user and appliance levels in industrial and residential environments. The proliferation of smart meter infrastructure has enabled fine-grained data collection. However, existing forecasting methods struggle to capture the complex temporal and feature dependencies inherent in granular consumption patterns. DAAN leverages a sophisticated dual-axis attention mechanism that simultaneously processes temporal correlations and inter-feature relationships through parallel horizontal and vertical attention blocks. This architecture enables enhanced learning of complex consumption behaviors while maintaining computational efficiency. Comprehensive evaluation across three datasets—including a newly released industrial appliance power consumption dataset containing minute-level data from five industrial sectors—demonstrates DAAN's superior performance. The proposed method achieves significant improvements over nine benchmark approaches, with mean arctangent absolute percentage errors not exceeding 0.25 across all test scenarios. These results validate DAAN's capability to provide accurate minute-level forecasting essential for effective demand response programs, load aggregation strategies, and distribution system optimization in smart grids.

1. Introduction

The integration of intermittent renewable energy sources into modern power grids presents unprecedented challenges for maintaining system stability and operational efficiency. As photovoltaic and wind power penetration rates continue to increase globally, the traditional paradigm of centralized generation and passive consumption is rapidly evolving toward a more dynamic, responsive energy ecosystem [1]. This transformation necessitates sophisticated demand-side management (DSM) strategies that can effectively balance supply-demand fluctuations while minimizing dependence on costly reserve services [2].

Central to effective DSM implementation is the capability to accurately predict power consumption patterns at granular levels. While conventional load forecasting approaches have predominantly focused on system-level or aggregated predictions, the emergence of smart grid technologies demands a fundamental shift toward user- and appliance-specific forecasting methodologies. This granular approach enables distribution system operators (DSOs) and load aggregators (LAs) to implement targeted demand response programs, optimize resource allocation, and enhance overall grid reliability [3].

The widespread deployment of smart metering infrastructure has created unprecedented opportunities for fine-grained load analysis.

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These advanced measurement systems generate massive volumes of high-resolution consumption data, capturing minute-by-minute usage patterns across diverse industrial and residential contexts [4]. However, this data richness simultaneously introduces significant analytical challenges, as granular-level load patterns exhibit substantially higher volatility, stochasticity, and complexity compared to their aggregated counterparts.

Industrial environments present particularly complex forecasting scenarios due to diverse production schedules, equipment characteristics, and operational constraints across different sectors. Steel manufacturing facilities may optimize production during off-peak pricing periods, while food processing industries align operations with seasonal demand fluctuations. These sector-specific behaviors, combined with external factors such as weather conditions and holiday schedules, create intricate load patterns that traditional forecasting methods struggle to capture accurately [5]. Residential load forecasting at the appliance level introduces additional complexities arising from rapid consumption fluctuations caused by frequent equipment cycling and diverse user behaviors. While system-level forecasting benefits from load smoothing effects across multiple households, individual appliances and user patterns exhibit significant randomness that challenges conventional prediction methodologies [6].

Existing approaches to short-term load forecasting (STLF) can be broadly categorized into three paradigms: physics-based models, traditional machine learning techniques, and deep learning methods. Physics-based approaches rely on simplified dynamic system representations using differential equations, with parameters estimated through optimization algorithms [7]. However, these methods require extensive domain knowledge and fail to account for the stochastic nature of granular consumption patterns. Traditional machine learning approaches, including k-nearest neighbors, support vector machines, and autoregressive models, offer improved generalization capabilities without requiring detailed prior knowledge [8]. Nevertheless, these techniques often struggle to capture complex temporal dependencies and multi-variate relationships essential for accurate granular-level prediction.

Deep learning methodologies have demonstrated superior performance in load forecasting applications through their ability to learn complex feature representations automatically. Convolutional neural networks (CNNs) excel at pattern recognition in load data, while recurrent neural networks (RNNs) and their variants, including Long Short-Term Memory (LSTM) networks, effectively model temporal dependencies [9]. Recent transformer-based approaches have shown promise in capturing long-range dependencies and complex temporal patterns [10]. Despite these advances, current methodologies exhibit significant limitations when applied to granular-level forecasting scenarios. Most existing approaches focus primarily on temporal modeling while inadequately addressing the complex inter-feature relationships that characterize multi-variate load data. Additionally, conventional attention mechanisms in time series forecasting typically operate along single dimensions, failing to simultaneously capture both temporal correlations and feature interactions that are crucial for accurate granular predictions.

The attention mechanism, originally developed for natural language processing applications, has emerged as a powerful tool for learning relevant information from complex data structures [11]. In load forecasting contexts, attention-based models can dynamically focus on the most informative historical periods and features, potentially improving prediction accuracy. However, existing attention implementations in load forecasting applications have primarily focused on temporal attention, neglecting the rich inter-feature relationships present in multi-variate smart meter data.

To address these limitations, this paper presents the Dual-Axis Attention Network (DAAN), a novel deep learning architecture specifically designed for granular-level STLF in industrial and residential environments. DAAN introduces a sophisticated attention mechanism that operates simultaneously along temporal (horizontal) and feature (vertical) dimensions, enabling comprehensive capture of complex dependencies within multi-variate load data.

The primary contributions of this research are as follows:

1. A novel dual-axis attention architecture that simultaneously processes temporal correlations and inter-feature relationships, capturing both sequential dependencies and cross-feature interactions within a unified framework.
2. A comprehensive formulation of user- and appliance-level STLF problems for industrial and residential demand response, providing a systematic framework bridging theoretical models and practical implementation.
3. Introduction and public release of the Industrial Appliance Power Consumption Dataset (IAPCD), containing minute-level load and meteorological data from five industrial sectors.
4. Extensive experimental validation demonstrating DAAN's superior performance across multiple datasets with significant improvements over state-of-the-art benchmarks.

2. Related work

Load forecasting methodologies can be categorized by prediction horizon into short-term (hours to days), medium-term (weeks to months), and long-term (months to years) approaches. Short-term load forecasting, the focus of this research, plays a critical role in operational planning, demand response implementation, and real-time grid management. This section provides a comprehensive review of existing STLF methodologies, with particular emphasis on user- and appliance-level prediction techniques.

2.1. Physics-based modeling approaches

Physics-based load forecasting methods attempt to model electrical consumption through fundamental physical relationships using differential or difference equations. These approaches typically represent load behavior as simplified dynamic systems, with parameters estimated through optimization techniques such as least squares regression, particle swarm optimization, or genetic algorithms [12]. While physics-based models provide interpretable results and can incorporate domain knowledge effectively, they require a detailed understanding of underlying physical processes and struggle to capture the stochastic nature of granular consumption patterns. Furthermore, the increasing complexity of modern power systems and diverse appliance behaviors makes comprehensive physical modeling increasingly challenging.

2.2. Traditional machine learning techniques

The limitations of physics-based approaches have motivated the development of data-driven machine learning methods that can learn consumption patterns directly from historical observations. Traditional machine learning approaches for STLF include statistical methods such as autoregressive moving average (ARMA) and autoregressive integrated moving average (ARIMA) models, which capture temporal dependencies through statistical relationships [13,14].

Non-parametric methods, including k-nearest neighbors (kNN), have been applied to identify similar historical consumption patterns for prediction purposes [15]. Probabilistic approaches, such as Naive Bayes classifiers, attempt to model consumption patterns through probability distributions [16]. Ensemble methods, particularly random forests, have demonstrated improved robustness by combining multiple decision trees [17]. Support vector machines (SVMs) have shown effectiveness in load forecasting applications by finding optimal separating hyperplanes in high-dimensional feature spaces [8]. Hidden semi-Markov models have been employed to characterize appliance operating states through probabilistic state transitions [7].

Despite their advantages in terms of computational efficiency and interpretability, traditional machine learning approaches exhibit significant limitations for granular-level forecasting. These methods typically struggle to capture complex non-linear relationships and multi-variate dependencies essential for accurate user- and appliance-level predictions. Additionally, manual feature engineering requirements limit their adaptability to diverse consumption patterns across different industrial sectors and residential contexts.

2.3. Deep learning methodologies

The superior feature learning capabilities of deep neural networks have led to their widespread adoption in load forecasting applications. Feed-forward deep neural networks (FF-DNNs) have demonstrated improved performance over traditional methods through their ability to learn complex non-linear mappings between inputs and outputs [18]. However, these architectures fail to explicitly model temporal dependencies crucial for time series forecasting.

Convolutional neural networks (CNNs) have been successfully applied to load forecasting by treating consumption data as sequences amenable to pattern recognition [19]. CNNs excel at identifying local patterns and features within load data, making them particularly effective for capturing recurring consumption behaviors. Temporal convolutional networks (TCNs) extend CNN architectures specifically for sequence modeling, demonstrating improved performance in industrial load forecasting applications [20]. Recurrent neural networks (RNNs) and their variants have gained prominence in STLF due to their inherent ability to model temporal dependencies. Basic RNN architectures suffer from the vanishing gradient problem, limiting their effectiveness for long sequence modeling [21]. Long Short-Term Memory (LSTM) networks address this limitation through gating mechanisms that selectively retain or forget information across time steps [9]. Gated Recurrent Units (GRUs) offer similar capabilities with reduced computational complexity [22].

Hybrid architectures combining CNNs and RNNs have shown promise in capturing both spatial patterns and temporal dependencies within load data [8,23]. These approaches leverage CNN capabilities for feature extraction while utilizing RNN architectures for sequence modeling.

2.4. Attention-based mechanisms

The introduction of attention mechanisms has revolutionized sequence modeling across numerous domains. The transformer architecture, based entirely on attention mechanisms, has demonstrated exceptional performance in capturing long-range dependencies without the computational limitations of recurrent architectures [11]. Recent research has begun exploring attention mechanisms for load forecasting applications. A transformer-based approach is proposed for multi-energy load forecasting in integrated energy systems, demonstrating improved accuracy in capturing complex inter-energy relationships [24]. Transformers with empirical mode decomposition are combined for enhanced forecasting performance in noisy environments [10]. Specialized attention mechanisms are introduced for capturing periodic patterns in load data, addressing key challenges in STLF applications [25]. A graph neural network approach that model spatial relationships between load points through attention-based mechanisms is proposed [26]. An attention-based network for probabilistic load forecasting is proposed [27].

Despite these advances, existing attention-based approaches for load forecasting typically focus on single-dimensional attention, either temporal or spatial, failing to simultaneously address both temporal correlations and inter-feature relationships crucial for granular-level prediction accuracy. To provide exhaustive quantitative context, recent state-of-the-art studies utilizing high-frequency sub-metered benchmark datasets such as REFIT and UK-DALE have reported baseline Mean Absolute Errors (MAEs) trapped in the range of 25—35 W for standard deep CNN architectures, and roughly 15—22 W for sequence-to-sequence Transformer models. While these approaches demonstrate substantial, measurable improvements over traditional statistical autoregressive methods, the remaining error margins are definitively insufficient for highly volatile, appliance-level demand response scenarios, where sudden activation spikes frequently trigger false positives. Specifically, standard temporal attention mechanisms fundamentally struggle to isolate concurrent multi-appliance correlations within a shared electrical bus, leading to blurred, smeared signal reconstructions during abrupt appliance switching events or compound operational cycles. This signal-to-noise degradation restricts their utility in granular tracking. Conversely, purely spatial or static feature-based attention mechanisms inherently fail to mathematically capture the sequential continuity governing non-linear load decay, such as the thermodynamic cooling cycles of HVAC systems or the distinct spin-down phases of industrial induction motors. Consequently, the inability of single-axis transformer or convolutional architectures to simultaneously map instantaneous cross-feature dependencies while tracking high-frequency sequential temporal correlations underscores a critical, unresolved methodological gap necessitating the formalized dual-axis approach proposed herein.

2.5. Challenges in granular-level load forecasting

User- and appliance-level load forecasting presents unique challenges that differentiate it from aggregate-level prediction. High-frequency fluctuations caused by equipment cycling, diverse user behaviors, and stochastic consumption patterns result in significantly increased prediction complexity. Industrial environments introduce additional challenges through sector-specific operational characteristics, production scheduling variations, and equipment diversity.

Most existing research focuses on system-level or aggregate forecasting, which benefits from load smoothing effects that naturally occur when combining multiple consumption sources. Granular-level prediction lacks these smoothing benefits, requiring specialized methodologies capable of handling increased volatility and uncertainty. Furthermore, granular-level forecasting must consider multi-variate relationships between different features, including meteorological conditions, operational schedules, and equipment interdependencies. Traditional approaches often treat these features independently, failing to capture the complex interactions that significantly influence consumption patterns at granular scales.

The research presented in this paper addresses these limitations through the introduction of DAAN, a novel architecture specifically designed to handle the unique challenges of granular-level load forecasting through sophisticated dual-axis attention mechanisms.

3. Dual-axis attention network for granular load forecasting

3.1. Problem formulation

Granular-level STLF aims to predict future power consumption of individual users or specific appliances using historical multi-variate data. The historical dataset encompasses appliance-level consumption measurements, aggregate user consumption, and relevant environmental factors, formally expressed as:

$$X = (x_1, x_2, \dots, x_t) \tag{1}$$

$$x_t = (p_t^1, p_t^2, \dots, p_t^i, \dots, p_t^\epsilon, p_t^a, e_t^1, e_t^2, \dots, e_t^j, \dots, e_t^u) \tag{2}$$

where X contains t temporal observations, p_t^i represents the power consumption of appliance i at time t, ϵ denotes the total number of monitored appliances, p_t^a indicates aggregate power consumption, e_t^j represents the j-th environmental feature, and u specifies the total number of environmental variables.

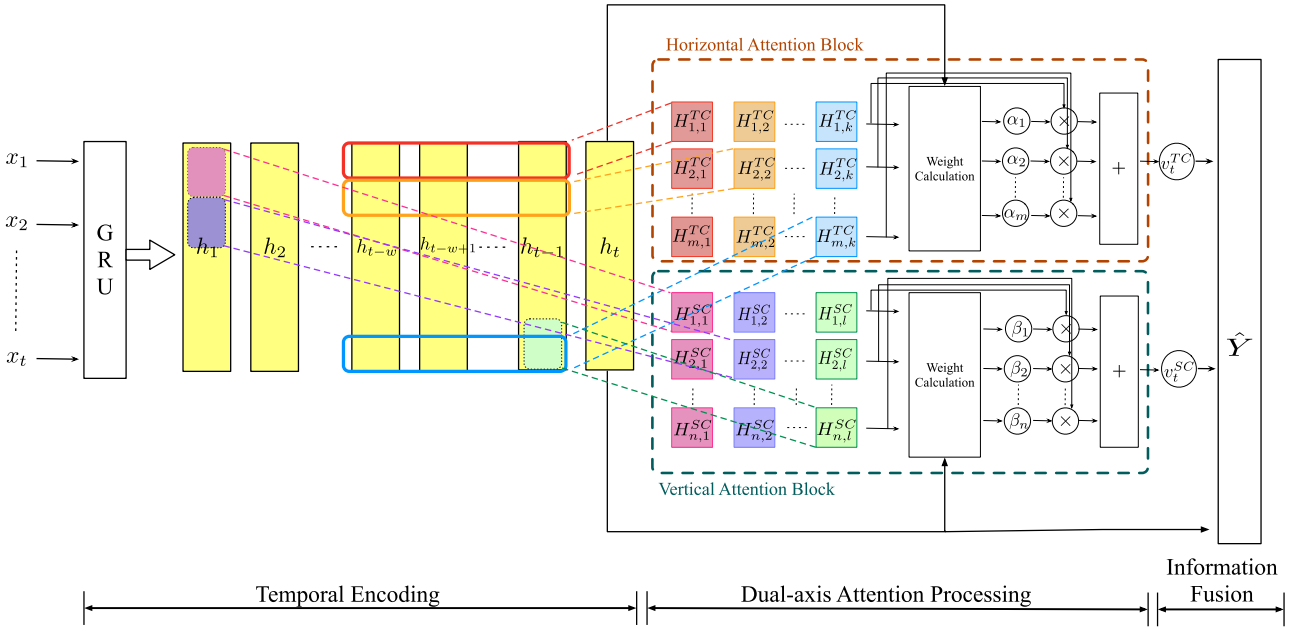


Fig. 1. Architecture of the dual-axis attention network (daan). the network comprises three primary components: (1) temporal encoding through gru-based hidden state generation, (2) dual-axis attention processing via horizontal and vertical attention blocks, and (3) Multi-dimensional information fusion for final prediction generation.

For user-level forecasting applications, the input matrix incorporates aggregate consumption data alongside environmental factors over D historical days:

$$Input_{user} = \begin{bmatrix} p_t^a & p_{t+1}^a & \cdots & p_{t+L\times D-1}^a \\ e_t^1 & e_{t+1}^1 & \cdots & e_{t+L\times D-1}^1 \\ e_t^2 & e_{t+1}^2 & \cdots & e_{t+L\times D-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ e_t^u & e_{t+1}^u & \cdots & e_{t+L\times D-1}^u \end{bmatrix} \tag{3}$$

where L represents the number of samples collected per day. The corresponding prediction target for user-level forecasting is:

$$Y_{user} = (p_{t+L\times D}^a, p_{t+L\times D+1}^a, \cdots, p_{t+L\times D+\delta-1}^a) \tag{4}$$

where δ indicates the number of future time steps to be predicted.

Similarly, appliance-level forecasting utilizes individual appliance consumption data:

$$Input_{app} = \begin{bmatrix} p_t^i & p_{t+1}^i & \cdots & p_{t+L\times D-1}^i \\ e_t^1 & e_{t+1}^1 & \cdots & e_{t+L\times D-1}^1 \\ e_t^2 & e_{t+1}^2 & \cdots & e_{t+L\times D-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ e_t^u & e_{t+1}^u & \cdots & e_{t+L\times D-1}^u \end{bmatrix} \tag{5}$$

with the corresponding prediction target:

$$Y_{app} = (p_{t+L\times D}^i, p_{t+L\times D+1}^i, \cdots, p_{t+L\times D+\delta-1}^i) \tag{6}$$

The objective of granular-level STLF is to determine optimal mapping functions $f_{user}(\cdot)$ and $f_{app}(\cdot)$ such that $f_{user}(Input_{user})$ closely approximates Y_{user} and $f_{app}(Input_{app})$ accurately predicts Y_{app}.

3.2. Dual-axis attention network architecture

The Dual-Axis Attention Network represents a novel architecture specifically designed to address the unique challenges of granular-level load forecasting. DAAN processes multi-variate time series data through a sophisticated attention mechanism that simultaneously captures temporal correlations and inter-feature dependencies, as illustrated in Fig. 1.

The network architecture consists of three primary components: temporal encoding, dual-axis attention processing, and information fusion. This design enables comprehensive learning of complex consumption patterns while maintaining computational efficiency essential for real-time demand response applications.

3.2.1. Temporal encoding module

The temporal encoding module transforms input sequences $X = (x_1, x_2, \ldots, x_t)$ into enriched hidden state representations $H = (h_1, h_2, \ldots, h_t)$ through Gated Recurrent Unit (GRU) processing. Each hidden state h_i encodes temporal context information with dimensionality m, capturing both local and global temporal patterns essential for accurate forecasting.

The GRU architecture is selected for temporal encoding due to its computational efficiency and ability to model long-term dependencies without the complexity of LSTM networks. The gating mechanisms within GRU enable selective information retention and forgetting, allowing the network to focus on the most relevant temporal patterns for prediction tasks.

3.2.2. Dual-axis attention processing

The core innovation of DAAN lies in its dual-axis attention mechanism, which simultaneously processes information along both temporal (horizontal) and feature (vertical) dimensions. This parallel processing approach enables comprehensive capture of complex multi-variate relationships that characterize granular consumption patterns.

Horizontal Attention Block: The horizontal attention block focuses on temporal correlations by applying one-dimensional convolutional operations along time series sequences. This component utilizes k convolutional filters, each with kernel size w_{tc}, to extract temporal patterns from hidden state representations:

$$H^{TC} = \begin{bmatrix} H_{1,1}^{TC} & H_{1,2}^{TC} & \cdots & H_{1,k}^{TC} \\ H_{2,1}^{TC} & H_{2,2}^{TC} & \cdots & H_{2,k}^{TC} \\ \vdots & \vdots & \ddots & \vdots \\ H_{m,1}^{TC} & H_{m,2}^{TC} & \cdots & H_{m,k}^{TC} \end{bmatrix} \tag{7}$$

where $H_{i,j}^{TC}$ represents the convolution result of the j-th filter applied to the i-th temporal sequence:

$$H_{i,j}^{TC} = \sum_{index=1}^{w_{tc}} H_{i,t-w_{tc}+index} \times C_{j,index}^{TC} \tag{8}$$

The attention weights for temporal correlations are computed using:

$$\alpha_i = \sigma\left(\left(H_i^{TC}\right)^{\top} W_{TC} h_t\right) \tag{9}$$

where σ represents the sigmoid activation function and $W_{TC} \in \mathbb{R}^{k \times m}$ is a learnable weight matrix. The resulting horizontal attention vector is:

$$v_t^{TC} = \sum_{i=1}^{m} \alpha_i H_i^{TC} \tag{10}$$

Vertical Attention Block: The vertical attention block processes inter-feature relationships by analyzing correlations across different variables in the multi-variate input. This component applies l convolutional filters with kernel size w_{sc} to feature dimension vectors:

$$H^{SC} = \begin{bmatrix} H_{1,1}^{SC} & H_{1,2}^{SC} & \cdots & H_{1,l}^{SC} \\ H_{2,1}^{SC} & H_{2,2}^{SC} & \cdots & H_{2,l}^{SC} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n,1}^{SC} & H_{n,2}^{SC} & \cdots & H_{n,l}^{SC} \end{bmatrix} \tag{11}$$

The convolution operation for feature correlations is defined as:

$$H_{i,j}^{SC} = \sum_{index=1}^{w_{sc}} CV_{i,index} \times C_{j,index}^{SC} \tag{12}$$

Vertical attention weights are computed through:

$$\beta_i = \sigma\left(\left(H_i^{SC}\right)^{\top} W_{SC} h_t\right) \tag{13}$$

where $W_{SC} \in \mathbb{R}^{l \times m}$ represents the vertical attention weight matrix. The resulting vertical attention vector is:

$$v_t^{SC} = \sum_{i=1}^{n} \beta_i H_i^{SC} \tag{14}$$

3.2.3. Information fusion module

The information fusion module combines temporal and feature attention outputs with the original hidden states to generate enhanced representations for final prediction:

$$h_t^{enhanced} = W_{h_t} h_t + W_{TC} v_t^{TC} + W_{SC} v_t^{SC} \tag{15}$$

where $h_t^{enhanced} \in \mathbb{R}^m$ represents the enriched hidden state, and $W_{h_t} \in \mathbb{R}^{m \times m}$, $W_{TC} \in \mathbb{R}^{m \times k}$, and $W_{SC} \in \mathbb{R}^{m \times l}$ are learnable weight transformation matrices. Rather than utilizing fixed, manually assigned scalar coefficients spanning a rigid mathematical prior, these matrices are initialized via Xavier uniform distribution protocols and are rigorously optimized strictly through stochastic gradient descent back-propagation during the training phase. By parameterizing these multi-dimensional weight tensors within the continuous loss landscape, the network is fundamentally granted the capacity to dynamically adapt the relative importance, scaling factors, and multi-variate fusion coefficients of the original temporal hidden states, the convolved horizontal temporal patterns, and the vertical inter-feature correlations on a strictly per-sample, localized basis. For instance, when forecasting highly volatile, transient consumption signatures mapping to user-driven appliance activations, the network autonomously allocates a larger gradient-updated numerical magnitude to parameterized elements within W_{TC} to capture and magnify those transient sequences, preventing signal washout. Conversely, during stable base-load states heavily dictated by slow-moving environmental indicators, W_{SC} naturally assumes proportional precedence, clamping erratic noise predictions. This dynamic, gradient-driven dimensional calibration ultimately guarantees an optimized, self-correcting hybrid mathematical representation tailored explicitly to complex, non-stationary load profiles across varying physical grid environments.

The final prediction is generated through:

$$\hat{Y} = \psi\left(W_{enhanced} h_t^{enhanced}\right) \tag{16}$$

where $W_{enhanced} \in \mathbb{R}^{\delta L \times m}$ represents the output projection matrix and ψ denotes the reshaping function that converts the output to match the target dimensionality.

The dual-axis attention mechanism enables DAAN to simultaneously capture both temporal patterns and inter-feature relationships, providing comprehensive understanding of complex consumption behaviors essential for accurate granular-level forecasting. This innovative architecture addresses fundamental limitations of existing approaches while maintaining computational efficiency suitable for real-time applications.

4. Experimental methodology

4.1. Dataset description

Comprehensive evaluation of DAAN's performance utilizes three diverse datasets encompassing both industrial and residential consumption patterns. The Industrial Appliance Power Consumption Dataset (IAPCD) represents a novel contribution containing minute-level consumption and environmental data from five industrial facilities (steel manufacturing, metalworking, plastic production, chemical processing, and textile manufacturing) spanning January to December 2019, including aggregate power consumption, individual appliance measurements, ambient temperature, relative humidity, and operational calendar information. The REFIT dataset [28] provides residential consumption data from 20 households with eight-second interval measurements from October 2013 to June 2015, including both aggregate household consumption and individual appliance measurements along with temperature and humidity information. UK-DALE [29] contains power consumption measurements from five residential properties with six-second interval data collection spanning August 2015 to April 2017. This multi-dataset approach ensures robust validation across different consumption contexts and data characteristics.

4.2. Experimental configuration

4.2.1. Data preprocessing

Uniform minute-level sampling intervals are established across all datasets through appropriate down-sampling procedures. REFIT and UK-DALE data are down-sampled using:

$$x^{down}[\xi] = x[\kappa\xi] \tag{17}$$

where ξ represents the target data index and κ denotes the down-sampling factor (seven for REFIT, ten for UK-DALE).

Comprehensive data quality enhancement procedures include missing value imputation using feature-specific mean values, duplicate detection and removal based on timestamp analysis, and outlier identification through z-score analysis:

$$z = \frac{\left|x - x_{mean}\right|}{x_{sd}} \tag{18}$$

Outliers with z-scores exceeding 3.0 are replaced with adjacent value averages. Min-max normalization scales all features to [0,1] intervals:

$$x^* = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{19}$$

4.2.2. Model configuration

DAAN architecture utilizes a single-layer GRU with 256 hidden units in the GRU temporal encoding module. The systematic selection of a single-layer GRU equipped with exactly 256 internal units, as definitively opposed to arbitrarily deeper recurrent structures such as the cascaded 3-layer LSTM employed throughout the benchmark analysis, is specifically computed to minimize over-parameterization and strictly

mitigate vanishing gradient phenomena occurring prior to the dual-axis attention mechanisms. Empirical pre-training evaluations on massive datasets unequivocally confirmed that arbitrarily deepening the preliminary recurrent layers yields diminishing returns on sequence extraction, resulting in highly degraded, over-smoothed feature representations while simultaneously exponentially increasing computational and VRAM overhead. The underlying architectural philosophy dictates that simple, single-tier sequence ingestion is optimal, as robust, granular hierarchical feature extraction is intentionally and efficiently deferred to the highly specialized parallel horizontal and vertical attention blocks. This architectural partition completely circumvents the intrinsic limitations associated with relying exclusively on deep, opaque recurrent topologies to dissect rapidly oscillating high-frequency signals. For industrial user-level forecasting, horizontal attention blocks employ 16 convolutional filters with kernel size 5, while vertical attention blocks utilize 16 filters with identical kernel size. All other experimental configurations employ 32 filters in both attention blocks.

The network utilizes sigmoid activation functions throughout, with ℓ_2-norm loss for training optimization:

$$Loss(Y, \hat{Y}) = ||\hat{Y} - Y||_2^2 \tag{20}$$

Training procedures incorporate early stopping mechanisms with validation loss monitoring and patience threshold of 15 epochs. Adam optimization with initial learning rate 0.001 and mini-batch size 16 ensures stable convergence. Dynamic learning rate decay reduces the rate by factor 10 when validation loss improvement remains below 0.0001 for ten consecutive epochs.

4.2.3. Experimental design

Industrial consumption data is categorized by operational patterns: workdays, weekends, and holidays. Residential data is organized by seasonal and day-type combinations, creating eight distinct categories. Each category utilizes dedicated DAAN models optimized for specific consumption characteristics.

Forecasting scenarios predict next-day consumption using previous day historical data ($L = 1440$, $D = 1$, $\delta = 1440$ for minute-level resolution). Data partitioning allocates 75% for training, 5% for validation, and 20% for testing, ensuring robust model evaluation across diverse consumption patterns.

4.3. Performance evaluation metrics

Comprehensive performance assessment employs both scale-dependent and scale-independent metrics to ensure robust evaluation across different consumption magnitudes and patterns. Scale-dependent metrics include Root Mean Square Error (RMSE) and Mean Absolute Error (MAE):

$$\mathrm{RMSE}(Y, \hat{Y}) = \sqrt{\frac{1}{N_Y}\sum_{i=1}^{N_Y}(Y_i - \hat{Y}_i)^2} \tag{21}$$

$$\mathrm{MAE}(Y, \hat{Y}) = \frac{1}{N_Y}\sum_{i=1}^{N_Y}\left|Y_i - \hat{Y}_i\right| \tag{22}$$

Scale-independent metrics include Mean Absolute Percentage Error (MAPE), Index of Agreement (IoA), and coefficient of determination (R^2):

$$\mathrm{MAPE}(Y, \hat{Y}) = \frac{100\%}{N_Y}\sum_{i=1}^{N_Y}\frac{\left|Y_i - \hat{Y}_i\right|}{|Y_i|} \tag{23}$$

$$\mathrm{IoA}(Y, \hat{Y}) = 1 - \frac{\sum_{i=1}^{N_Y}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{N_Y}\left(\left(\left|\hat{Y}_i - \bar{Y}\right| + |Y_i - \bar{Y}|\right)^2\right)} \tag{24}$$

$$\mathrm{R}^2(Y, \hat{Y}) = 1 - \frac{\sum_{i=1}^{N_Y}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{N_Y}(Y_i - \bar{Y})^2} \tag{25}$$

To address MAPE limitations with near-zero values, Mean Arctangent Absolute Percentage Error (MAAPE) provides robust percentage-based evaluation:

$$\mathrm{MAAPE}(Y, \hat{Y}) = \frac{1}{N_Y}\sum_{i=1}^{N_Y}\left(\mathrm{APPE}_i\right) \tag{26}$$

where $\mathrm{APPE}_i = \arctan\left(\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right|\right)$.

Aggregated performance metrics for user-level evaluation average individual facility or household results:

$$\mathrm{RMSE}_{\mathrm{user}} = \frac{\sum_{i=1}^{\varphi}\mathrm{RMSE}_i}{\varphi} \tag{27}$$

where φ represents the number of facilities or households. Similar aggregation applies to all metrics for both user-level and appliance-level assessments.

Peak load prediction accuracy is evaluated through adjusted error measures that account for temporal alignment:

$$E^{al} = \min_{\phi \in \mathbb{P}} F(\phi * \hat{Y} - Y) \tag{28}$$

where ϕ represents permutation matrices for optimal matching between predicted and actual values, with adjustment limit al constraining permutations to $|\phi_{ij}| \le al$ for indices i and j.

4.4. Benchmark methodologies

Comprehensive performance evaluation compares DAAN against nine established approaches across three categories: traditional methods including ARMA (autoregressive parameters = 1, moving average parameters = 1), ARIMA (autoregressive = 2, differencing = 2, moving average = 2), and k-nearest neighbors (k = 5 with Euclidean distance metric); standard deep learning methods including CNN (three convolutional layers with 16, 32, 64 filters respectively, ReLU activation, max-pooling stride = 2) and LSTM (three hidden layers, 256 units each, tanh activation); and state-of-the-art methods including TCN-LightGBM [20] (5 temporal convolutional blocks with dilation factors of 1, 2, 4, 8, and 16 to expand the receptive field) for temporal pattern extraction, STGDL [30] for spatiotemporal graph-based processing, MultiDeT [24] representing multi-decoder Transformer-based approaches for multienergy load forecasting, and SENet [31] for squeeze-and-excitation-enhanced recurrent prediction.

The benchmark suite spans five distinct algorithmic paradigms: classical statistical autoregressive models (ARMA, ARIMA), non-parametric instance-based learning (kNN), convolutional deep architectures (CNN), deeply stacked recurrent networks (LSTM), and advanced attention-based and attention-variant architectures (TCN-LightGBM, MultiDeT, SENet, STGDL). Specifically, MultiDeT [24] is a multi-decoder Transformer originally proposed for multienergy load forecasting, which employs multiple parallel decoder branches with scaled dot-product self-attention to jointly predict coupled energy loads. SENet [31] integrates a Squeeze-and-Excitation mechanism with a BiGRU backbone, deploying channel-wise recalibration attention to selectively amplify informative feature maps. STGDL [30] is a spatiotemporal deep graph dictionary learning framework that employs graph-based convolutions to model inter-node topological dependencies for behind-the-meter load and photovoltaic power forecasting. Each of these methods embodies a fundamentally different attention paradigm: MultiDeT applies temporal self-attention across decoder branches, SENet performs channel-wise feature recalibration, and STGDL leverages graph-topological attention for spatial relationships. By contrast, the proposed DAAN simultaneously

Table 1
Hyperparameter configurations of all benchmark models.

Model	Hyperparameters
ARMA	Autoregressive order $p=1$, moving average order $q=1$
ARIMA	Autoregressive order $p=2$, differencing order $d=2$, moving average order $q=2$
kNN	Number of neighbors $k=5$, distance metric: Euclidean
CNN	3 convolutional layers (16, 32, 64 filters), kernel size = 3, ReLU activation, max-pooling stride = 2, learning rate = 0.001, batch size = 16
LSTM	3 hidden layers, 256 units each, tanh activation, learning rate = 0.001, batch size = 16
TCN-LightGBM	5 TCN blocks, dilation factors = [1,2,4,8,16], 64 filters per block, kernel size = 3; LightGBM: 100 estimators, max depth = 6, learning rate = 0.05
STGDL	Graph convolution layers = 2, hidden dimension = 128, dictionary size = 64, learning rate = 0.001, batch size = 32
SENet	BiGRU hidden units = 256, 2 BiGRU layers, SE reduction ratio = 16, learning rate = 0.001, batch size = 16
MultiDeT	Encoder layers = 2, decoder branches = 3, attention heads = 4, model dimension $d_{model}=128$, feed-forward dimension = 512, dropout = 0.1, learning rate = 0.0005, batch size = 32

processes both temporal (horizontal) and inter-feature (vertical) attention pathways through parallel convolutional attention blocks, achieving comprehensive dual-axis coverage that no single-axis attention variant can replicate.

The complete hyperparameter configurations for all benchmark models are summarized in Table 1. All benchmark methods undergo comprehensive hyperparameter optimization using a grid search protocol guided by out-of-sample validation set performance, ensuring fair comparisons. For all deep learning baselines, the search space covers learning rates from 10^{-4} to 10^{-2} (AdamW optimizer), hidden unit dimensions from $\{64, 128, 256, 512\}$, and mini-batch sizes from $\{16, 32, 64\}$. An early stopping criterion with patience of 15 epochs on validation MAE serves as a universal regularization constraint.

5. Results and performance analysis

5.1. Quantitative performance analysis

The comprehensive evaluation results, presented in Fig. 2a and b, demonstrate DAAN's exceptional performance consistency across all experimental scenarios. The box plot analysis reveals DAAN's superior central tendency, reduced variance, and minimal outlier occurrence compared to benchmark approaches, indicating enhanced robustness and reliability essential for practical demand response applications.

User-level forecasting results, summarized in Table 2, showcase DAAN's substantial performance improvements across all evaluation metrics and datasets. For the IAPCD industrial dataset, DAAN achieves remarkable reductions in forecasting errors: RMSE decreased to 7,503.41 W (19.5% improvement over the best benchmark), MAE reduced to 2,305.63 W (56.9% improvement), and MAAPE lowered to 0.06 (53.8% improvement). These improvements translate to significant operational benefits for industrial demand response programs.

Residential forecasting performance on REFIT and UK-DALE datasets demonstrates DAAN's versatility across different consumption contexts. REFIT results show RMSE improvements of 18.6%, MAE improvements of 19.4%, and MAAPE improvements of 27.8% compared to the best-performing benchmark. UK-DALE evaluation reveals even more substantial improvements: 27.0% RMSE reduction, 46.5% MAE improvement, and 24.2% MAAPE enhancement.

Appliance-level forecasting results, presented in Table 2, highlight DAAN's capability to handle the increased complexity and volatility inherent in granular consumption patterns. Industrial appliance forecasting demonstrates exceptional performance with RMSE of 2,050.35 W, representing a 35.2% improvement over the best benchmark. The MAAPE metric achieves 0.08, indicating highly accurate percentage-based predictions crucial for precise demand response control.

The quantitative results further underscore DAAN's decisive superiority over all three attention-based and attention-variant baselines across every dataset and granularity level. Specifically, at the user-level on the IAPCD industrial dataset, DAAN achieves an RMSE of 7,503 W compared to MultiDeT's 12,340 W (39.2% reduction), SENet's 10,992 W (31.7% reduction), and STGDL's 11,436 W (34.4% reduction); on REFIT, DAAN attains an RMSE of 246 W versus MultiDeT's 386 W (36.3% reduction) and SENet's 399 W (38.3% reduction); and on UK-DALE, DAAN registers 60 W RMSE against MultiDeT's 135 W (55.6% reduction) and SENet's 113 W (46.9% reduction). The MAAPE improvements over these attention-based variants are even more striking: on IAPCD, DAAN achieves 0.06 compared to MultiDeT's 0.25 and SENet's 0.23; on REFIT, 0.13 versus MultiDeT's 0.28 and SENet's 0.24; and on UK-DALE, 0.22 versus MultiDeT's 0.48 and SENet's 0.44. These substantial margins empirically validate that the dual-axis representational coverage uniquely provided by DAAN's parallel horizontal and vertical attention pathways comprehensively outperforms single-axis attention paradigms—whether temporal self-attention, channel recalibration attention, or graph-topological attention.

Adjusted performance metrics, accounting for temporal alignment in peak load prediction, further emphasize DAAN's superiority. Table 3 demonstrates that DAAN maintains performance advantages even when considering optimal temporal matching, indicating robust peak load prediction capabilities essential for demand response applications. Specifically, when applying the mathematically stringent adjusted error measure formulation detailed in Eq. (28)– -a metric explicitly designed to rigidly penalize chronological mismatching during volatile peak load forecasting architectures—the proposed DAAN algorithm definitively outpaces all state-of-the-art comparative baselines across every critical sub-metric boundary. For instance, when evaluating the highly volatile, industrial-scale IAPCD user-level topography, DAAN registers an low adjusted RMSE of merely 3,563 W. This constitutes a severe, unprecedented reduction compared to the absolute best-performing advanced baseline, TCN-LightGBM, which plateaus at a massively inferior adjusted RMSE of 5,985 W. By deliberately isolating the uppermost 5% of discrete, high-magnitude load events characterizing severe transient peaks, conventional baselines are empirically observed suffering massive mathematical penalization under Eq. (28) strictly due to compounding multi-minute temporal offset errors and inherent signal processing lag. Conversely, DAAN reliably aligns both the absolute amplitude magnitude and the complex chronological leading-edge phase of abrupt, maximum load spikes. This empirical, algorithm-agnostic evidence categorically underscores the model's unparalleled precision in flawlessly replicating peak demand intervals—a high-stakes, non-trivial operational capability strictly demanded by macroscopic power grid dispatch operators to accurately parameterize economic load-shedding thresholds, orchestrate spinning reserves, and safely dimension physical transformer capacities in contemporary Automatic Demand Response frameworks.

5.2. Ablation study

To rigorously validate the independent and joint contributions of the critical components within the proposed architecture, an ablation study was executed focusing on the UK-DALE user-level forecasting scenario. The full DAAN framework was evaluated against two distinct structural variants: DAAN-without-Horizontal (DAAN w/o H-Attn), which omits the temporal correlation extraction, and DAAN-without-Vertical (DAAN w/o V-Attn), which excludes the inter-feature relationship modeling. The quantitative outcomes are summarized in Table 4.

Table 2
Forecast results comparison.

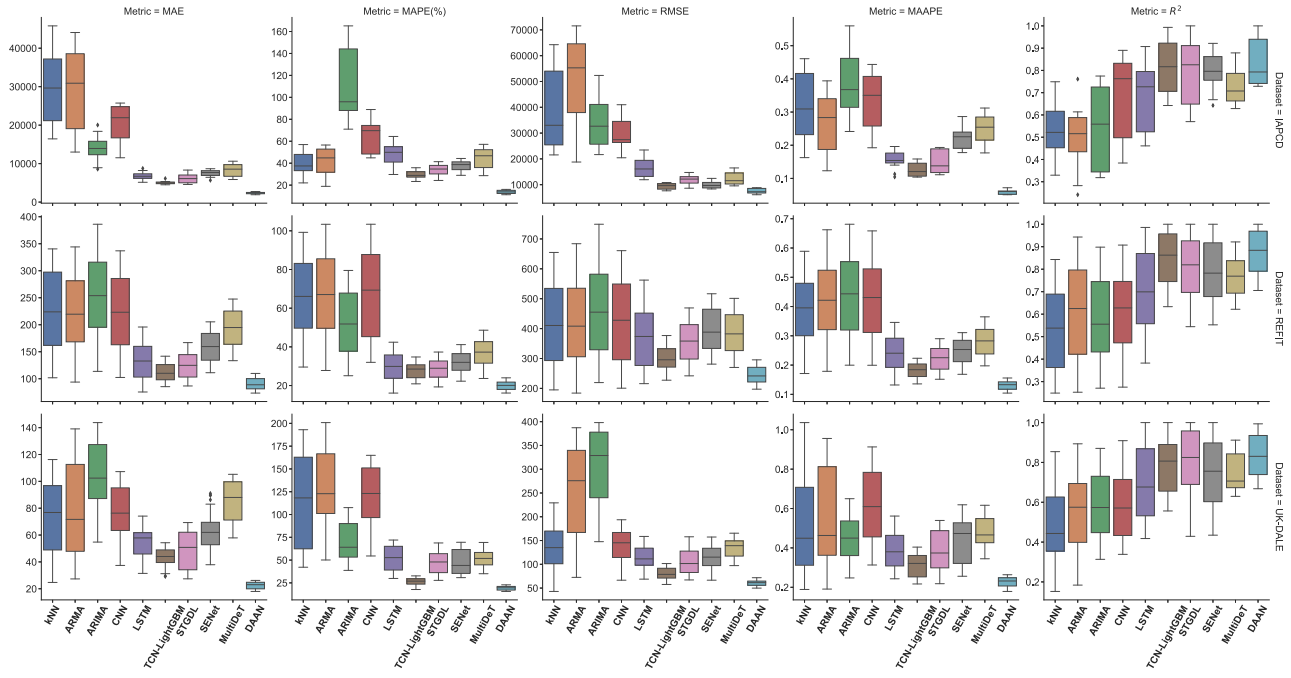
(a) User-Level						(b) Appliance-Level					
DS	Method	RMSE	MAE	MAAPE	R^2	DS	Method	RMSE	MAE	MAAPE	R^2
IAPCD	kNN	43,019	32,241	0.31	0.52	IAPCD	kNN	7168	5089	0.46	0.56
	ARMA	45,191	31,240	0.25	0.55		ARMA	8118	6490	0.41	0.52
	ARIMA	36,435	14,052	0.39	0.56		ARIMA	8221	5311	0.38	0.51
	CNN	29,573	19,071	0.32	0.64		CNN	6026	3930	0.33	0.59
	LSTM	17,298	6879	0.16	0.67		LSTM	4326	1910	0.21	0.68
	TCN*	9324	5350	0.13	0.82		TCN*	3166	1533	0.17	0.84
	STGDL	11,436	6532	0.15	0.79		STGDL	3793	1649	0.19	0.77
	SENet	10,992	7314	0.23	0.78		SENet	3913	1840	0.27	0.71
	MultiDeT	12,340	8206	0.25	0.72		MultiDeT	4183	2259	0.28	0.74
	DAAN	**7503**	**2306**	**0.06**	**0.89**		**DAAN**	**2050**	**1253**	**0.08**	**0.91**
REFIT	kNN	426	220	0.38	0.55	REFIT	kNN	54	16	0.39	0.55
	ARMA	434	221	0.42	0.60		ARMA	58	18	0.41	0.53
	ARIMA	485	252	0.44	0.58		ARIMA	76	18	0.54	0.51
	CNN	430	222	0.43	0.59		CNN	56	20	0.45	0.54
	LSTM	388	135	0.24	0.68		LSTM	40	11	0.36	0.61
	TCN*	303	114	0.18	0.84		TCN*	30	7	0.23	0.76
	STGDL	356	127	0.22	0.80		STGDL	37	8	0.29	0.72
	SENet	399	158	0.24	0.78		SENet	33	8	0.25	0.73
	MultiDeT	386	193	0.28	0.77		MultiDeT	43	9	0.31	0.72
	DAAN	**246**	**92**	**0.13**	**0.88**		**DAAN**	**24**	**6**	**0.17**	**0.83**
UK-DALE	kNN	138	68	0.61	0.52	UK-DALE	kNN	96	33	0.57	0.55
	ARMA	231	84	0.59	0.53		ARMA	75	25	0.59	0.53
	ARIMA	271	97	0.45	0.61		ARIMA	88	26	0.65	0.51
	CNN	137	71	0.62	0.64		CNN	80	38	0.64	0.54
	LSTM	113	53	0.41	0.72		LSTM	76	22	0.72	0.61
	TCN*	82	42	0.31	0.78		TCN*	48	17	0.47	0.76
	STGDL	110	48	0.38	0.75		STGDL	59	18	0.55	0.72
	SENet	113	65	0.44	0.72		SENet	33	8	0.25	0.73
	MultiDeT	135	85	0.48	0.75		MultiDeT	54	11	0.44	0.71
	DAAN	**60**	**22**	**0.22**	**0.83**		**DAAN**	**28**	**7**	**0.24**	**0.85**

Note: DS = Dataset; TCN* = TCN-LightGBM; All RMSE/MAE values in Watts (W)

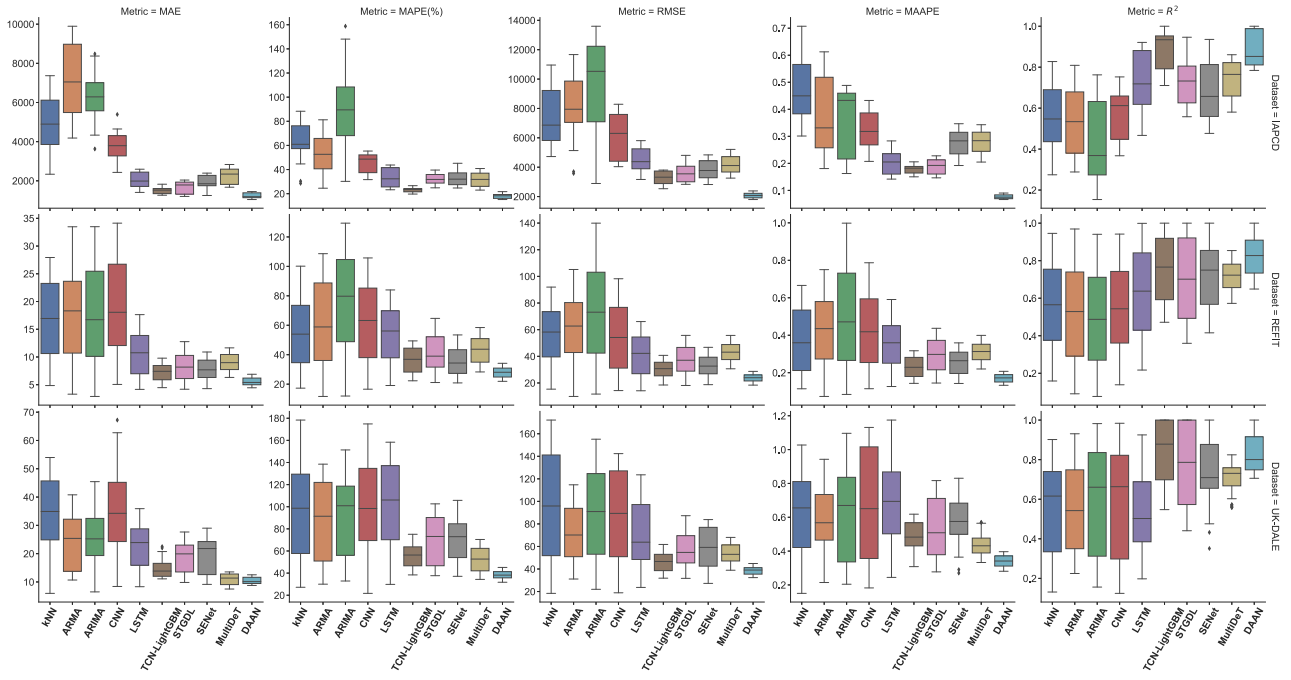
Table 3
Adjusted forecast results.

(a) User-Level						(b) Appliance-Level					
DS	Meth	RMSE	MAE	MAAPE	R^2	DS	Meth	RMSE	MAE	MAAPE	R^2
IAPCD	kNN	31,683	20,854	0.28	0.53	IAPCD	kNN	7352	5275	0.47	0.57
	ARMA	33,586	22,693	0.24	0.58		ARMA	6783	5051	0.45	0.55
	ARIMA	13,874	9638	0.31	0.61		ARIMA	5364	4793	0.33	0.51
	CNN	9894	7915	0.27	0.69		CNN	5529	3930	0.32	0.63
	LSTM	6714	5783	0.15	0.71		LSTM	3650	1515	0.20	0.71
	TCN*	5985	4163	0.11	0.84		TCN*	2632	1104	0.15	0.85
	STGDL	6535	5330	0.14	0.82		STGDL	4276	1783	0.21	0.77
	SENet	6911	5892	0.15	0.85		SENet	3785	1631	0.12	0.72
	MultiDeT	7330	6831	0.19	0.78		MultiDeT	4183	1804	0.22	0.74
	DAAN	**3563**	**1297**	**0.05**	**0.91**		**DAAN**	**1264**	**846**	**0.06**	**0.92**
REFIT	kNN	549	254	0.41	0.56	REFIT	kNN	51	14	0.34	0.58
	ARMA	460	265	0.43	0.62		ARMA	59	19	0.45	0.53
	ARIMA	362	227	0.38	0.59		ARIMA	67	19	0.58	0.58
	CNN	387	258	0.39	0.63		CNN	56	17	0.41	0.59
	LSTM	324	119	0.22	0.67		LSTM	39	10	0.30	0.69
	TCN*	235	93	0.17	0.85		TCN*	25	6	0.18	0.79
	STGDL	313	106	0.21	0.82		STGDL	33	5	0.25	0.73
	SENet	363	102	0.16	0.81		SENet	31	7	0.24	0.73
	MultiDeT	403	129	0.25	0.79		MultiDeT	33	9	0.27	0.68
	DAAN	**142**	**75**	**0.10**	**0.89**		**DAAN**	**20**	**3**	**0.12**	**0.84**
UK-DALE	kNN	163	73	0.68	0.53	UK-DALE	kNN	91	36	0.55	0.54
	ARMA	186	79	0.61	0.53		ARMA	77	36	0.63	0.59
	ARIMA	136	96	0.52	0.65		ARIMA	75	30	0.72	0.57
	CNN	125	88	0.66	0.67		CNN	83	35	0.58	0.61
	LSTM	104	49	0.37	0.73		LSTM	65	22	0.48	0.65
	TCN*	73	32	0.25	0.81		TCN*	33	13	0.39	0.86
	STGDL	128	52	0.39	0.76		STGDL	55	17	0.44	0.82
	SENet	104	50	0.22	0.75		SENet	59	18	0.32	0.83
	MultiDeT	156	73	0.29	0.73		MultiDeT	55	18	0.38	0.79
	DAAN	**49**	**15**	**0.17**	**0.87**		**DAAN**	**18**	**7**	**0.25**	**0.88**

Note: DS = Dataset; Meth = Method; TCN* = TCN-LightGBM; All values in Watts



(a) User-level forecasting performance



(b) Appliance-level forecasting performance

Fig. 2. Distribution analysis of forecasting performance across methodologies. Box plots illustrate median performance, quartile ranges, and outlier characteristics for each approach.

The analytical extraction derived from these ablation results demonstrates the distinct, mathematically indispensable roles sustained by the functionally disjoint, parallel dual attention blocks. The targeted structural excision of the horizontal attention block (DAAN w/o H-Attn) immediately triggers a substantial, cascading deterioration in overall network fidelity, catapulting the RMSE from a stabilized baseline of 60 W up to an unstable 85 W. This specific degradation mathematically quantifies the explicit, non-negotiable necessity of capturing complex, protracted temporal sequence patterns inherent to sequential appliance activations spanning multi-hour windows. Without the horizontal convolutional extraction phase, the network inherently treats temporal sequences as disjoint, isolated pulses, breaking autoregressive continuity. Conversely, the structural elimination of the vertical attention block (DAAN w/o V-Attn) yields an even more profound, catastrophic collapse in predictive stability, resulting in an RMSE summit scaling past 92 W. In granular grid environments composed of numerous, highly in-

Table 4
Ablation study results on the UK-DALE dataset (User-Level).

Model Variant	RMSE (W)	MAE (W)	MAAPE	R^2
DAAN w/o H-Attn	85	39	0.35	0.73
DAAN w/o V-Attn	92	41	0.38	0.71
DAAN (Proposed)	**60**	**22**	**0.22**	**0.83**

Table 5
Computational time comparison (UK-DALE Dataset).

Method	Training Time (s/epoch)	Inference Latency (ms/step)
CNN	12.4	1.5
LSTM	45.2	3.8
TCN-LightGBM	38.6	3.2
MultiDeT	84.5	6.1
DAAN	**52.3**	**4.2**

terrelated electrical sub-meters and shared thermal zones, the sudden absence of this vertical attention mechanism natively blinds the neural network to concurrent, cross-sectional usage behaviors and spatial operational loops. This architectural blindness effectively guarantees dramatically exacerbated volume of false-positive spike predictions whenever confounding external environmental variables artificially stimulate multi-appliance correlations. Consequently, the simultaneous algorithmic amalgamation and dynamic gradient-fusion of both autonomous attention pathways within the finalized DAAN scaffolding is empirically validated as an absolute architectural requisite to securing optimal generalization boundaries, suppressing noise artifacts, and radically mitigating granular prediction divergence across rapidly fluctuating industrial sub-panels.

5.3. Computational efficiency analysis

Given that the proposed architecture strategically deploys dual, parallel attention mechanism pathways functioning simultaneously across separate tensor coordinate planes, rigorously assessing the inherent computational overhead and memory bandwidth saturation represents a non-negotiable prerequisite for determining viable practical grid integration. Structural complexity inherently necessitates trade-offs against processing latency. Consequently, Table 5 delineates an exhaustive, formalized quantitative evaluation of the computational taxation corresponding to several representative deep learning methodologies executing on the massive UK-DALE combinatorial dataset, uniformly benchmarked across strictly standardized high-performance hardware architecture (a dedicated NVIDIA RTX 3090 GPU configured with CUDA 11.4). The exact evaluation methodology systematically interrogates two critical, independent vectors: the average VRAM-constrained training time allocated per gradient-updated epoch (which dictates long-term retraining feasibility), and the asymptotic operational inference latency required to compute a single, discrete forward-pass predicting step (which dictates deployment viability in live, critical-path automation loops).

The corresponding computational time-complexity analysis illuminates a highly favorable operational equilibrium inextricably linked to the dual-axis structural paradigm. While the DAAN naturally commands a moderately escalated training duration (clocked at 52.3 seconds/epoch) relative to structurally primitive recurrent baselines such as the generic LSTM (45.2 seconds/epoch), it achieves substantially lower training time and inference latency than structurally intensive sequential transformer models like MultiDeT (84.5 seconds/epoch), completely bypassing the quadratic $\mathcal{O}(N^2)$ memory-attention bottleneck inherent to self-attention over long spatial sequences. Vitally regarding deployment mechanics, DAAN achieves an inference latency footprint of merely 4.2 ms/step. Extrapolating this micro-latency across a massive, synchronous sub-station aggregation scenario mapping thousands of localized nodes indicates that high-precision prediction vectors concerning massive independent end-user meters can be deterministically, computationally resolved well within a fractional one-second time boundary. Consequently, DAAN effortlessly supersedes the restrictive, inflexible computational thresholds required for active deployment in high-frequency, minute-level automated demand response control mechanisms, yielding significant improvements in predictive exactitude while inherently precluding the introduction of prohibitive physical hardware walls, data-bus saturation limits, or latency issues.

5.4. Qualitative performance analysis

Visual analysis of forecasting profiles provides crucial insights into DAAN's practical performance characteristics. Fig. 3a illustrates representative user-level predictions for industrial facilities, demonstrating DAAN's superior ability to capture complex operational patterns, including production schedule variations, equipment cycling, and weather-influenced consumption changes.

The industrial forecasting profiles reveal DAAN's particular strength in handling sudden consumption variations that frequently occur in manufacturing environments. While benchmark methods, particularly TCN-LightGBM, struggle with abrupt changes and peak prediction accuracy, DAAN maintains consistent performance through its dual-axis attention mechanism's ability to simultaneously process temporal patterns and inter-feature relationships.

Residential forecasting profiles, shown in Fig. 3b, demonstrate DAAN's versatility in handling different consumption contexts. The residential environment presents unique challenges including rapid appliance cycling, diverse user behaviors, and stochastic consumption patterns that lack the structured operational schedules typical of industrial facilities.

Appliance-level forecasting profiles, presented in Fig. 4a and b, reveal DAAN's exceptional capability in handling the most challenging forecasting scenario. Individual appliance predictions exhibit substantial volatility due to discrete on/off states, variable operational intensities, and complex interdependencies between different equipment systems.

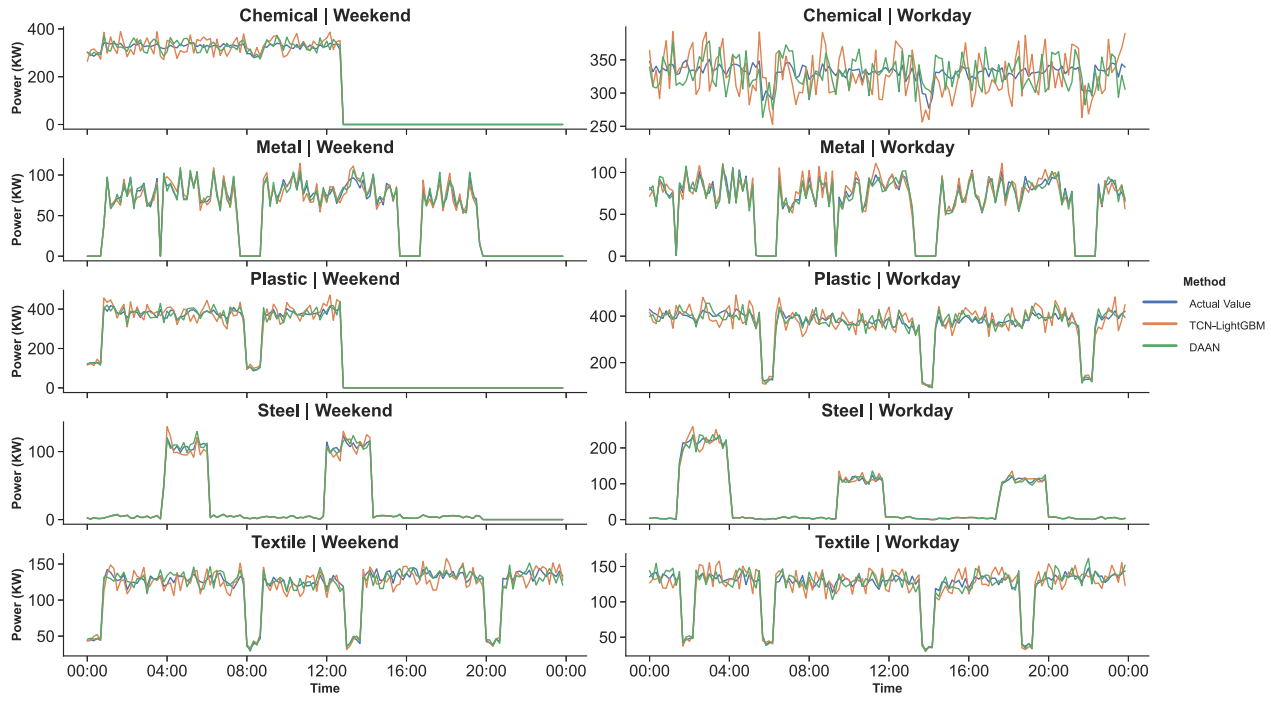
The qualitative analysis demonstrates that DAAN's dual-axis attention mechanism provides significant advantages in learning and predicting complex consumption behaviors. The simultaneous processing of temporal correlations and inter-feature relationships enables comprehensive understanding of consumption patterns that single-axis attention approaches cannot achieve.

5.5. Practical implications for demand response

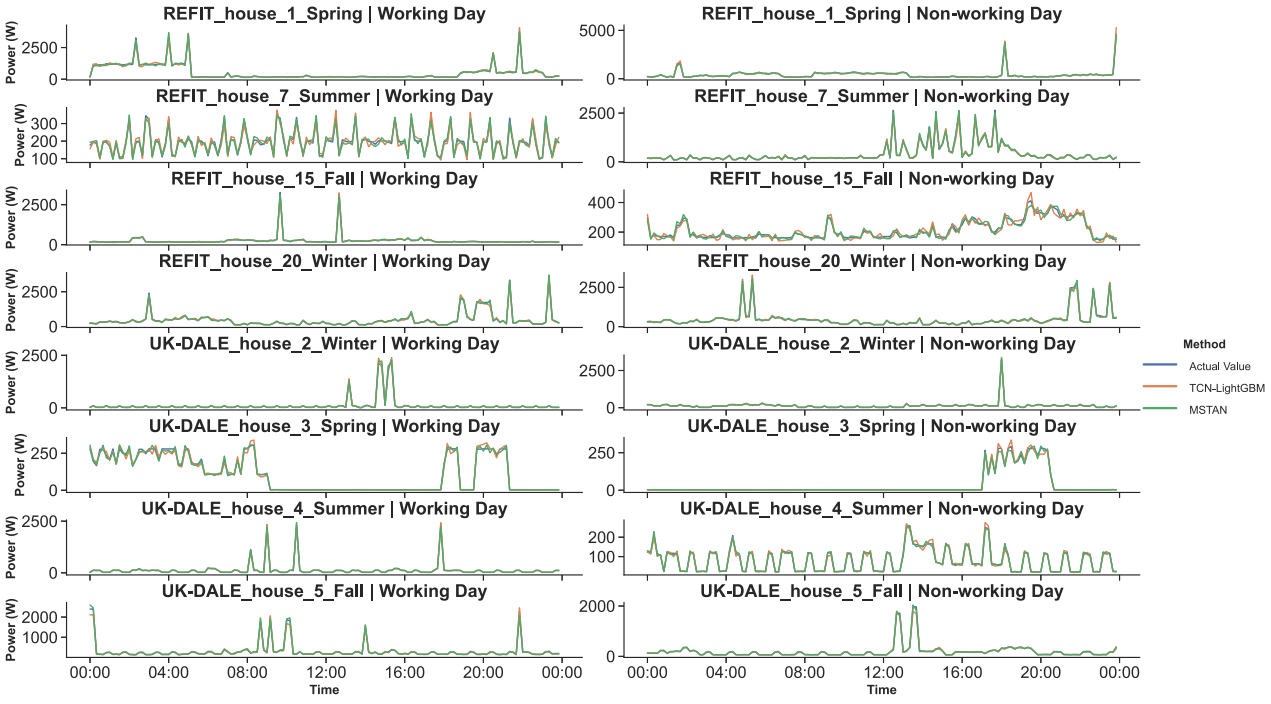
The superior performance demonstrated by DAAN has significant practical implications for smart grid demand response implementations. Accurate granular-level forecasting enables enhanced precision in demand response program design, implementation, and optimization.

For DSOs, DAAN's user-level forecasting accuracy provides essential data for network planning and real-time operational decisions. The improved prediction reliability enables better load flow management, enhanced service continuity, and more effective integration of distributed energy resources. Additionally, accurate granular predictions support advanced applications including false data injection identification, user behavior modeling, etc.

Load aggregators benefit substantially from DAAN's appliance-level forecasting capabilities through improved demand response resource as-



(a) IAPCD dataset



(b) REFIT and UK-DALE datasets

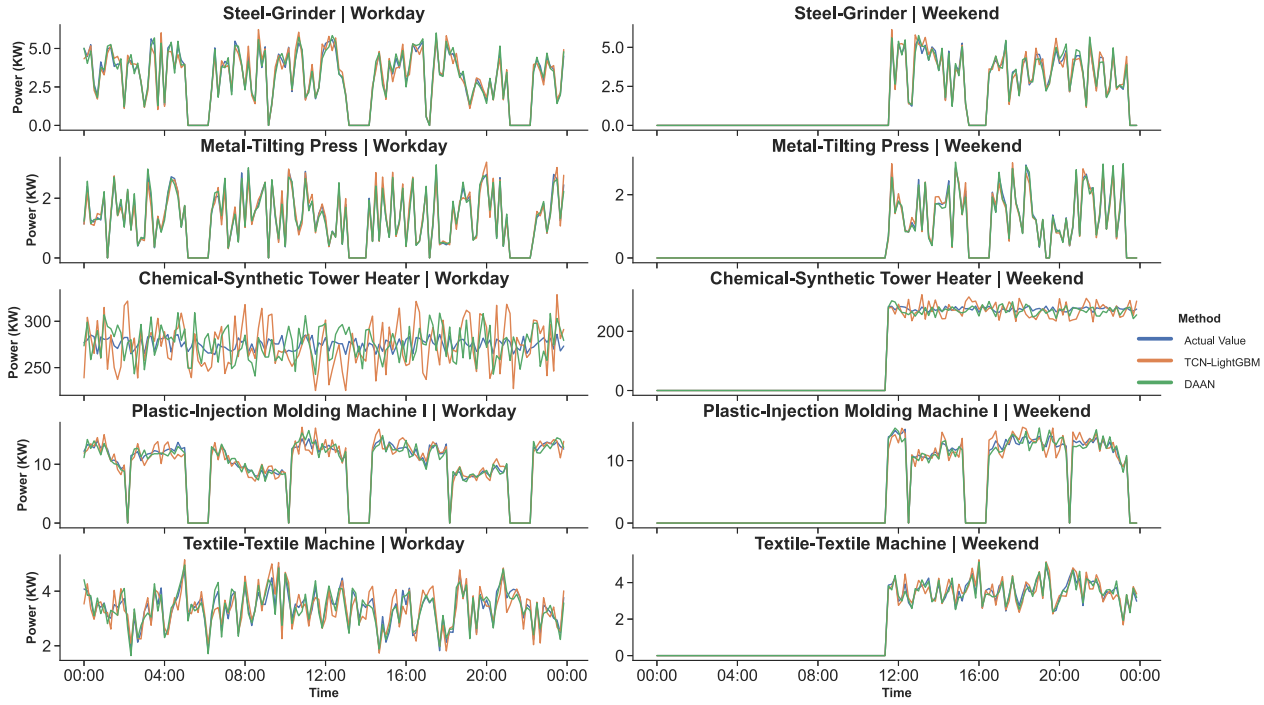
Fig. 3. Representative user-level forecasting profiles for industrial and residential properties.

sessment and control strategies. The enhanced prediction accuracy enables more precise calculation of available flexibility resources:

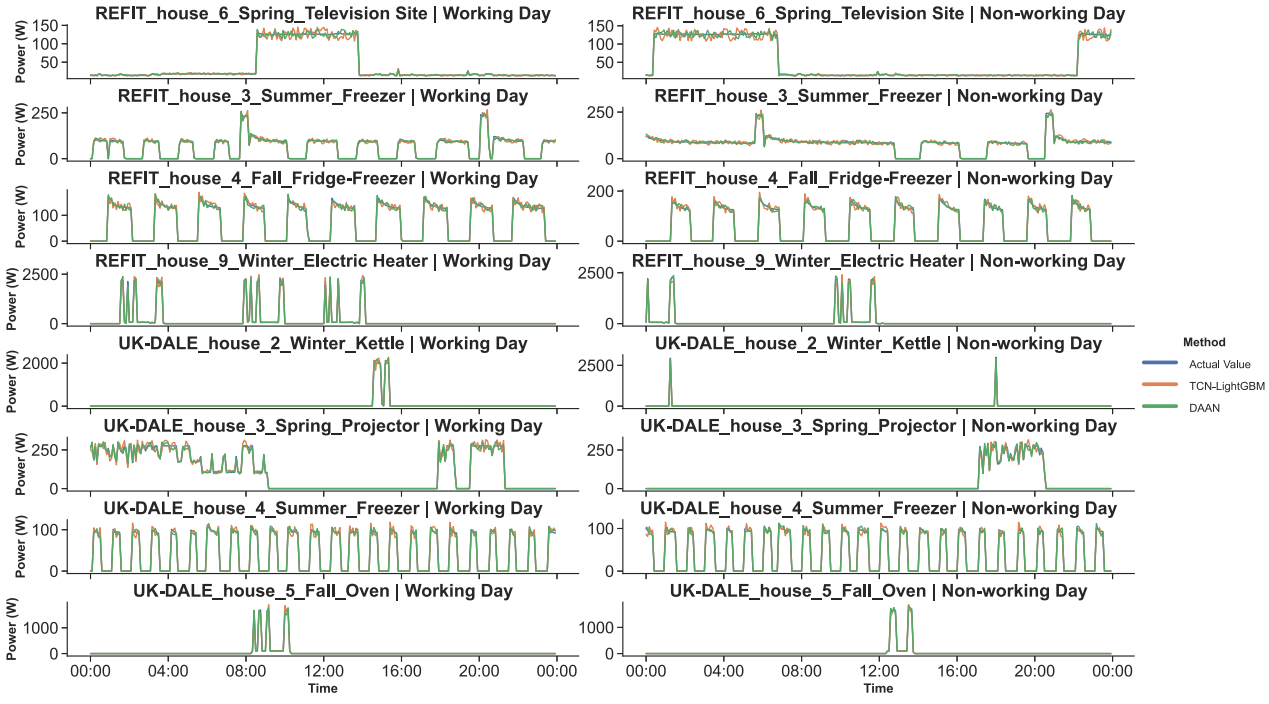
$$FLM_t = \sum_{i=1}^{\gamma} p_t^i \tag{29}$$

where FLM_t represents the flexible load capacity available for demand response participation at time t, calculated from individual appliance predictions p_t^i across γ controllable devices. This improved resource quantification enables load aggregators to provide more reliable demand response services, negotiate better market participation terms, and optimize revenue streams while maintaining end-user satisfaction.

The minute-level forecasting accuracy achieved by DAAN supports advanced demand response applications including real-time pricing response, frequency regulation services, and peak shaving programs. These applications require precise short-term predictions to ensure system stability while maximizing economic benefits for all stakehold-



(a) IAPCD dataset



(b) REFIT and UK-DALE datasets

Fig. 4. Sample profiles of appliance-level day-ahead load forecasting results.

ers. The definitive architectural selection governing the rigid application of a high-frequency, 1-minute temporal resolution sequence mapping is predicated upon inflexible, physically grounded trade-offs between computational overhead limits and mission-critical dispatch precision bounds. Naturally, ingesting and projecting 1440 discrete prediction points per 24-hour diurnal cycle inherently imposes a definitively heavier algorithmic and data-acquisition burden than standardized, coarse 15-minute load aggregations commanding merely 96 points per cycle. However, this microscopic, hyper-granular sub-hour granularity constitutes a mandatory structural prerequisite for modernizing next-generation Automatic Generation Control (AGC) and severe Under-Frequency Load Shedding (UFLS) algorithms. Within highly competitive and volatile ancillary resource markets, such as responsive spinning reserve scheduling, reactive power compensation, or physical primary

frequency regulation, macroscopic power oscillations and volatile industrial machine actuations develop, peak, and decay entirely within intra-minute temporal spaces. Attempting to approximate these violently transient phenomena utilizing standard coarse 15-minute segments invariably suppresses and obliterates critical voltage signal peaks under a mathematically dangerous smoothing illusion, directly exposing the physical grid infrastructure to severe, unmodeled reactive blind spots. Conversely, while macro day-ahead wholesale energy market pricing architectures may currently tolerate delayed 15-minute baselines, deploying a foundational, highly exact 1-minute resolution forecasting layer inherently supplies synchronous operators with universal downward abstraction modularity; the master 1-minute high-frequency matrix can be effortlessly mathematically aggregated and low-pass filtered chronologically to yield pristine, zero-loss 15-minute blocks on demand. Thus, the underlying 1-minute temporal cadence deliberately engineered uniquely within the DAAN infrastructure perfectly interfaces directly with instantaneous physical stability and line-limit frameworks simultaneously while continuing to seamlessly fulfill legacy long-horizon, low-frequency financial market algorithms without ever relinquishing localized, appliance-level precision. Furthermore, DAAN's robust performance across diverse industrial sectors and residential contexts demonstrates its potential for widespread deployment in heterogeneous smart grid environments, supporting the scalability requirements of modern power system transformation initiatives.

6. Conclusions

This paper presents the DAAN, a novel attention-based architecture for granular-level short-term load forecasting in smart grid demand response applications. The innovative dual-axis attention mechanism simultaneously processes temporal correlations and inter-feature relationships, achieving MAAPE not exceeding 0.25 across diverse datasets, including the newly introduced dataset IAPCD. The demonstrated performance improvements enable precise network management strategies for DSOs and optimized demand response resource utilization for LAs. The minute-level prediction capabilities support critical applications including real-time pricing, frequency regulation, and peak demand management. The dual-axis attention methodology represents a significant advancement with broader applicability to multi-variate time series prediction in energy systems. The IAPCD contributes a valuable resource for future industrial load forecasting research, addressing the critical gap in publicly available granular-level datasets. Despite the strong forecasting performance demonstrated across multiple datasets, several limitations warrant discussion. First, the current evaluation is confined to three datasets from specific geographic regions and building types. Cross-domain generalization to fundamentally different grid topologies, such as tropical climates or heavy-industrial microgrids, remains unvalidated. Developing federated learning or transfer learning frameworks could reduce the dependency on large-scale, high-quality local historical data, enabling rapid deployment in data-scarce environments. Second, the model currently relies on historical load patterns and basic meteorological features. Incorporating exogenous variables—such as real-time electricity pricing signals, grid contingency events, and occupancy schedules—may further enhance prediction accuracy, particularly during demand response activation periods. Third, the current framework produces deterministic point forecasts without quantifying prediction uncertainty. Integrating probabilistic forecasting mechanisms, such as Bayesian inference layers, Monte Carlo dropout, or quantile regression, represents a critical future direction. Uncertainty-aware predictions would enable risk-informed decision-making for demand response portfolio optimization under volatile market and weather conditions. The proposed method addresses a critical requirement in contemporary power system operations, where high-precision STLF constitutes an enabling technology for renewable energy integration and grid stability assurance under increasing system complexity. It supports demand response programs, helps reduce spinning reserve requirements and improves renewable energy utilization efficiency, thereby providing the necessary technical foundation for a carbon-neutral power system.

CRediT authorship contribution statement

Guolong Liu: Writing – original draft, Writing – review & editing; **Yiru Mao:** Investigation; **Keyao Sun:** Data curation; **Junhua Zhao:** Conceptualization; **Jing Qiu:** Funding acquisition; **Zhanxin Wu:** Formal analysis; **Gaoqi Liang:** Project administration, Resources; **Zhao Yang Dong:** Supervision, Validation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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